

Systematic Literature Review: Effective Prompting in AI Systems

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Abstract

This systematic literature review examines what makes prompting effective in large language models, drawing on 14 sources across natural language processing and HCI research. Beyond mapping the research landscape, the review aims to offer practical guidance for everyday AI users looking to get more out of their interactions. The central finding is that NLP and HCI research are measuring different things: NLP assumes users arrive with clear, fixed goals, while HCI shows they usually don't. Prompt strategy effects are real but task-dependent, and no strategy fully overcomes the structural limitations of the underlying model. The users who get the most out of AI are not necessarily the best prompt writers but the ones who iterate, question outputs, and bring critical thinking to the whole interaction. There is a ceiling that neither better prompting nor better habits can fully close, and future research has meaningful work to do in understanding it.

Introduction

AI tools like ChatGPT have become a significant part of everyday life; according to a 2025 Pew Research Center survey, 27% of U.S. adults interact with AI on a daily or near-daily basis (McClain et al., 2025). Many of those who use it rely on a simple approach: asking one model one question and expecting the right answer. However, research has shown that outputs from AI can differ significantly based on task type, prompt design, model choice, and how iteratively you engage across exchanges (Godoy, 2025; Chen et al., 2026; Pyo et al., 2026; Jo et al., 2026).

A key challenge is that there is often a mismatch between what a user means and what the model actually interprets (Jo et al., 2026). This goes deeper than prompt phrasing alone; users frequently come to these systems with goals that are still forming, and models are not well-equipped to bridge that gap. Because of this, a growing area of research known as prompt engineering has emerged to study how different ways of phrasing instructions can improve model behavior.

In this systematic literature review, I will look at this topic from two connected but methodologically distinct perspectives: natural language processing (NLP) and human-computer interaction (HCI). NLP research tends to measure prompting effectiveness through performance on controlled benchmark tasks, treating the prompt as a relatively fixed input to be optimized. HCI research, on the other hand, captures a messier and more realistic picture: one where users iterate across multiple prompts, switch between models, and adapt their strategies based on output. These two fields are asking related questions but measuring success very differently, and that gap is itself revealing. Beyond academic performance, prompting also has real consequences in high-stakes domains like software engineering, healthcare, and professional decision-making, where poorly structured prompts can lead to unreliable or even harmful outputs. This review draws on 14 sources across NLP and HCI research, examining prompting effectiveness in both controlled benchmark settings and real-world use.

The questions guiding this review are:

- 1. How do different prompt engineering strategies affect model performance, output alignment, and reliability across task types?**

- 2. How do findings from NLP benchmark research and HCI user studies diverge in their accounts of what makes prompting effective?**
- 3. What does existing research suggest about how users should approach prompting in practice?**
- 4. To what extent is prompting a social and communicative act, and what does this imply for how humans and AI systems should interact beyond single-shot prompt design?**

This review finds that while NLP and HCI research both give insight into effective prompting, they measure it in very different ways. That gap has real effects on how users should think about AI interaction, both at the level of individual prompts and in terms of how they approach AI use as a whole.

Methods

Sources were identified primarily through arXiv, with additional sources retrieved via Google Scholar and IEEE Xplore. The following searches were conducted:

- "Prompt engineering methods" for NLP survey literature
- "Prompt engineering human response benchmark" for NLP-oriented examples
- "Prompt engineering human response benchmark" for HCI-oriented examples
- "Human-AI interaction prompt behavior" for HCI research
- "How users verify AI outputs" for additional HCI research

In total, 14 sources were identified and reviewed, categorized as either NLP-oriented or HCI-oriented based on their primary methodology. NLP sources tend to measure prompting effectiveness through controlled benchmarks and performance metrics, while HCI sources tend to use qualitative or behavioral methods with real users in naturalistic settings. Some sources did not fall cleanly on one side and were assigned based on whichever methodology most closely reflected their primary contribution. This distinction is not always clear in the literature itself, and that ambiguity is part of what this review examines.

Sources were then mapped to the four research questions guiding this review, ranked from most to least central (see Figure 1 below).

Limitations

This review only encountered sources testing AI on coding, reasoning, and reading tasks. Many other application domains exist but could not be included given the scope of the assignment. Additionally, arXiv skews toward recent and NLP-heavy work, which may have influenced the balance of sources toward technical prompting research over HCI studies. It is also worth noting that 14 sources is a small sample by SLR standards, and a more exhaustive search would likely surface additional findings, particularly on the HCI side.

Figure 1: Sources matched to RQs (see Annotated Bibliography for findings on each source)

Title of Source	RQs the source answers (most to least)
One Is Not Enough: How People Use Multiple AI Models in Everyday Life	Q2, Q4
A Survey of Prompt Engineering Methods in Large Language Models for Different NLP Tasks	Q1, Q2
Atlas of Human-AI Interaction (v1): An Interactive Meta-Science Platform for Large-Scale Research Literature Sensemaking	Q4
Alvorada-Bench: Can Language Models Solve Brazilian University Entrance Exams?	Q3, Q1
Rethinking Autonomy: Preventing Failures in AI-Driven Software Engineering	Q4, Q3
Prompt Optimization with Human Annotations Using Pairwise Comparison	Q1
Optimizing Prompts for Large Language Models: A Causal Approach	Q1, Q2
Alignment has a Fantasia Problem	Q3, Q2, Q4
Localized Calibrated Uncertainty in Code Language Models	Q2, Q4
Understanding Critical Thinking in Generative Artificial Intelligence Use: Development, Validation, and Correlates of the Critical Thinking in AI Use Scale	Q3, Q4
Assessing the Feasibility of a Video-Based Conversational Chatbot Survey for Measuring Perceived Cycling Safety: A Pilot Study in New York City	Q4, Q3
Responsible AI in Construction Safety: Systematic Evaluation of Large Language Models and Prompt Engineering	Q1, Q4
The Impact of Response Latency and Task Type on Human-LLM Interaction and Perception	Q3, Q4
LLMs and People Both Learn to Form Conventions -- Just Not with Each Other	Q4

Background: Some Prompt Engineering Strategies

Prompt engineering refers to the practice of designing natural language instructions to elicit better responses from LLMs, operating entirely on the model's embedded knowledge without requiring retraining or fine-tuning (Vatsal & Dubey, 2024). At its simplest, prompting can be as direct as submitting a query with no additional structure, known as basic or zero-shot prompting, or it can include a small number of worked examples to guide the model, known as few-shot prompting. Beyond these baselines, researchers have developed increasingly structured strategies to improve performance on more demanding tasks. Chain-of-Thought (CoT) prompting, for instance, asks the model to reason through a problem step by step before arriving at an answer, mimicking how humans decompose complex problems. Self-Consistency builds on CoT by generating multiple reasoning paths and selecting the most consistent answer across them. Tree-of-Thoughts extends this further by having the model explore branching solution paths rather than committing to a single chain. These strategies form the core of what NLP research tests when evaluating prompting effectiveness, and understanding their differences is necessary context for the findings that follow.

Results

RQ1: How do different prompt engineering strategies affect model performance, output alignment, and reliability across task types?

The most comprehensive overview comes from Vatsal and Dubey (2024), who synthesize 44 papers covering 39 strategies across 29 NLP tasks. Key strategies examined include

Chain-of-Thought (CoT), Self-Consistency, Tree-of-Thoughts, Program-of-Thoughts, and Plan-and-Solve. The central finding is that no single strategy dominates universally: CoT produces up to 39% improvement over basic prompting on mathematical reasoning tasks, but gains shrink significantly elsewhere, and basic prompting remains competitive or best on some medical and translation tasks (Vatsal & Dubey, 2024). The performance gap between engineered and basic prompting is largest precisely where tasks require multi-step reasoning.

Godoy (2025) tests zero-shot, role-playing, and chain-of-thought across 270,900 responses from 20 models on Brazilian university entrance exams. Surprisingly, prompting strategy had minimal effect, typically under one percentage point of variance between strategies on the same model (Godoy, 2025). What drove performance differences was task type and model architecture, not prompt design.

Chen et al. (2026) compare static versus dynamic prompt selection across math, visualization, and data analytics benchmarks. Their proposed framework, Causal Prompt Optimization (CPO), finds that most existing optimization approaches confound prompt quality with query difficulty, and correcting for this through causal inference produces meaningful gains particularly on hard queries, with CPO achieving 82% accuracy compared to 77–80% for competing methods (Chen et al., 2026).

Lu & Lee (2024) evaluate prompt optimization across 26 generative tasks, comparing human pairwise feedback against numerical scoring. Their key finding is that prompt optimization benefits meaningfully from iterative human feedback, and that pairwise comparison is an efficient and reliable way to incorporate that feedback without requiring extensive annotation

overhead. Prompts were highly sensitive to phrasing across all 26 tasks, and performance improved consistently as feedback accumulated across iterations (Lu & Lee, 2024).

Sammour et al. (2024) evaluate Direct prompting, CoT, and Few-Shot prompting on standardized safety certification exams using GPT-3.5 and GPT-4o. Beyond overall accuracy figures of 73.8% and 84.6% respectively, the most striking finding is the breakdown of consistent failure modes across all strategies: 38% lack of knowledge, 31% reasoning flaws, 24% memory issues, and 7% calculation errors (Sammour et al., 2024). These failures persisted regardless of prompting strategy used, suggesting a ceiling effect that prompt engineering alone cannot overcome.

Core finding for RQ1

Across these five sources, prompt strategy effects vary by task rather than holding across the board; task type and model architecture consistently predict performance more reliably than prompt design does (Godoy, 2025; Vatsal & Dubey, 2024). Structured strategies like CoT and Tree-of-Thoughts meaningfully improve performance on multi-step reasoning tasks, but that advantage shrinks or disappears on tasks requiring factual recall, translation, or domain-specific knowledge. No prompting strategy fully makes up for the model's underlying weaknesses: reasoning flaws, knowledge gaps, and memory issues show up consistently regardless of how the prompt is written (Sammour et al., 2024). Prompt engineering is a useful but limited tool, and how much it helps depends heavily on what kind of work the task is actually asking the model to do.

RQ2: How do findings from NLP benchmark research and HCI user studies diverge in their accounts of what makes prompting effective?

HCI research paints a messier picture of AI interaction than benchmark studies suggest. Jo et al. (2026) argue that a core failure of current AI systems is what they call the Fantasia problem: models are optimized to respond to prompts as complete expressions of user intent, when in reality users frequently arrive with goals that are still forming, underspecified, or only partially articulated. This produces failures like premature execution, false satisfaction, and anchoring, outcomes that look helpful on the surface but are misaligned with what the user actually needed. Pyo et al. (2026) provide empirical grounding for this picture, finding that real users do not interact with a single model through a single well-crafted prompt. Instead, they build personal hierarchies across multiple models, assign different tools to different task stages, cross-check outputs for reliability, and iterate on their own prompts when results fall short.

NLP benchmark research, by contrast, operates under a fundamentally different set of assumptions. Vatsal and Dubey (2024) surveyed 39 prompting strategies across 29 NLP tasks, measuring effectiveness through accuracy on controlled datasets, a methodology that implicitly assumes user intent is fixed, clearly expressed, and fully captured in a single prompt. Chen et al. (2026) take this a step further by automating prompt selection through causal inference, producing a system that dynamically matches the best strategy to each query. CPO is sophisticated, but the user's intent is still treated as a stable input to be matched against an optimal prompt. Neither paper has a framework for users whose goals are still forming, who iterate across exchanges, or who switch models mid-task. Gros and Devanbu (2025) come

closest to acknowledging this gap, their paper on detecting where LLM-generated code deviates from user intent explicitly flags in its limitations section that how confidence signals should be displayed to users in interactive settings remains unaddressed future work. That acknowledgment is itself revealing: it marks the boundary where NLP benchmark research reaches the edge of what it can answer and stops short of the questions HCI research takes as its starting point.

Core finding for RQ2

NLP and HCI research are measuring different things and calling both "prompting effectiveness." NLP research focuses on optimizing a single prompt, assuming the user already knows what they want and just needs to say it clearly, an assumption that HCI research consistently shows real users do not satisfy. HCI research captures what NLP leaves out: that users often don't know exactly what they want until they've tried something, that they iterate, switch tools, and adjust based on how the model responds, and that the task itself and how hard it is shapes the whole interaction. The gap between these two fields is not just about methods, it's about what they think prompting actually is.

RQ3: What does existing research suggest about how users should approach prompting in practice?

One of the clearest practical takeaways from the research is that users should think about what kind of task they are doing before deciding how to prompt. Godoy (2025) finds that across 270,900 responses from 20 models, prompting strategy had almost no effect on performance, what mattered far more was the task type and which model was being used. Additionally, the

model confidence scores tracked actual accuracy well, meaning models can often signal when they are uncertain. Users should treat those signals as a cue to look more closely at an output rather than accepting it at face value.

Knowing where AI tends to fail is just as important as knowing how to prompt well. Sammour et al. (2024) evaluate six code generation models and find that every single one failed to meet basic safety thresholds, with failure modes breaking down as 38% lack of knowledge, 31% reasoning flaws, 24% memory issues, and 7% calculation errors, and these patterns held regardless of which prompting strategy was used. The practical implication is that users should not treat AI outputs as finished products, especially in high-stakes areas like software engineering or data analysis. The SAFE-AI framework proposed by Navneet & Chandra (2025) offers concrete guidance here: treat prompts as something to refine over time rather than fire off once, build in human oversight at key points, and use iterative feedback loops to catch errors before they compound.

Beyond strategy and oversight, the research points to something broader about how users should think about AI interaction overall. Jo et al. (2026) argue that one of the biggest mistakes users make is prompting too quickly, before they have a clear sense of what they actually want, often walking away satisfied with an output that does not really address their underlying need. This connects directly to what Lau et al. (2025) find in their critical thinking scale research: good AI use is not really about writing better prompts, it is about actively checking, questioning, and reflecting on what the model gives you rather than passively accepting it. The response latency study by Tan et al. (2026) adds another layer, finding that creation tasks naturally led users to iterate much more than advice tasks did, suggesting that complex or open-ended tasks are going to need multiple rounds of prompting, not just one. Ren et al. (2026) offer a real-world

illustration of this working well: in a conversational chatbot study measuring perceived cycling safety, conversational follow-up and clarity from users meaningfully improved the quality of the data collected.

Core finding for RQ3

Taken together, the research suggests that good AI use is less about finding the perfect prompt and more about how actively and thoughtfully you engage with the whole interaction. Users should match their approach to the task, pay attention to when the model signals uncertainty, know that certain kinds of failures like reasoning gaps and missing knowledge show up consistently, and treat AI outputs as a starting point to question rather than a final answer to accept. Additionally, reinforcing the technical side, even simple shifts from zero-shot to few-shot or chain-of-thought prompting can meaningfully improve results on reasoning-heavy tasks (Vatsal & Dubey, 2024), and these are strategies that don't require any special expertise to try.

RQ4: To what extent is prompting a social and communicative act, and what does this imply for how humans and AI systems should interact beyond single-shot prompt design?

A consistent finding across the research is that AI interaction works best when humans stay actively involved rather than stepping back and trusting the output. Gros and Devanbu (2025) show that even well-performing models produce outputs that deviate from user intent in ways that are not obvious without close inspection, and that confidence signals can help users know when to look more carefully. Sammour et al. (2024) reinforce this from a different angle, every

model they tested failed basic safety thresholds, with reasoning flaws and knowledge gaps showing up consistently regardless of how the prompt was written. Even in a domain where AI performed reasonably well overall, the failure modes were predictable and persistent enough that human judgment remained necessary throughout. The SAFE-AI framework that Navneet & Chandra (2025) propose responds to this directly, calling for human oversight built into the interaction process rather than added on after the fact. The implication across all three is the same: AI outputs are not finished products to be accepted, they are starting points to be evaluated.

The research also suggests that people already treat AI interaction as something closer to a conversation than a form submission, and that leaning into that framing tends to produce better results. Tan et al. (2026) find that users rated slower AI responses as more thoughtful and useful, applying the same social expectations to AI that they would to a human conversation partner where speed signals effort and care. Ren et al. (2026) find that in a real-world chatbot application examining NYC street cyclists on bike paths, conversational follow-up and engagement from users improved output quality meaningfully. Pyo et al. (2026) show that users who develop familiarity with a model over time, building accumulated context and a sense of the model's strengths, get more out of their interactions than users who treat every exchange as a fresh start.

That conversational engagement only pays off though if the user brings genuine critical thinking to the interaction. Lau et al. (2025) find that users who actively verify, question, and reflect on AI outputs get better results than those who accept outputs passively, and that this capacity for critical engagement is distinct from just knowing how to write a good prompt. Jo et al. (2026) add that AI systems are designed in a way that works against this, they respond confidently to whatever they receive, which rewards quick prompting and discourages the kind of reflection

that would actually improve outcomes. The burden falls on users to bring the thoughtfulness that the system won't prompt them for. As Ren et al. (2026) show in a simpler real-world context, what you put into the interaction shapes what you get out of it.

Even when users do all of this well, there is a ceiling that no amount of skill or effort can fully close. Jones et al. (2026) shows that humans and AI cannot build the kind of shared understanding that humans develop with each other, even when AI is explicitly prompted to behave more human-like, the lexical alignment and accuracy that characterize human-human communication remain out of reach. Archiwaranguprok et al. (2025) find through their mapping of over 1000 HCI papers that trust, explainability, and transparency are the dominant themes in human-AI interaction research, yet LLMs sit at the largest structural hole in the entire research network, meaning they are studied heavily across many domains but without any unified framework for how humans should actually interact with them. Even the field itself has not figured this out. The honest conclusion is that prompting is a social and communicative act, but one with real limits, AI is not a know-it-all, and no combination of prompt optimization and interaction strategy changes that.

Core finding for RQ4

Prompting is more like a conversation than a transaction, and the research consistently shows that treating it that way produces better outcomes. But effective human-AI interaction requires more than good prompting habits, it requires human judgment, active skepticism, and a realistic sense of what AI can and cannot do. At this point in time, AI systems have hard limits around reasoning, knowledge, and genuine shared understanding that persist regardless of how well a

user prompts. Prompting matters, but it is only one part of what good AI use looks like. Active engagement with the output is just as important as how you ask in the first place.

Discussion

The four research questions in this review were designed to be answered separately, but their findings point toward a single coherent picture. Prompt engineering, as studied in NLP research, is a real and useful tool, but it has clear limits. Across every benchmark study examined here, the gains from engineered prompting depended heavily on task type, shrank outside of multi-step reasoning contexts, and could not make up for the model's underlying weaknesses like reasoning flaws and knowledge gaps. The honest takeaway from the NLP side of this review is that prompt optimization helps, but not as much as the field sometimes implies.

Where NLP research runs out of answers, HCI research picks up something different. The users who got the most out of AI interaction in the studies examined here were not necessarily the ones who wrote the best prompts. They were the ones who iterated, cross-checked outputs, switched models when needed, brought genuine skepticism to what the model gave them, and treated the whole thing as an ongoing process rather than a single exchange. That pattern shows up consistently across the HCI sources, from the multi-model diary study to the critical thinking scale research to the response latency findings. Good AI use looks less like prompt optimization and more like active, thoughtful engagement with the whole interaction.

NLP research treats prompting as a technical problem: find the right strategy, apply it correctly, get better output. HCI research suggests the more important skill is something closer to critical thinking, knowing when to trust an output, when to question it, when to try again, and when to

switch tools entirely. These are not the same thing, and treating them as if they are leads to advice that does not actually help real users.

Even so, there is a ceiling that neither good prompting nor good interaction habits can fully close. The conventions study by Jones et al. (2026) is the clearest evidence of this: even when AI was explicitly prompted to behave more human-like, the kind of shared understanding that humans build with each other remained out of reach. Archiwaranguprok et al. (2025) find that despite enormous research attention, no unified framework exists for how humans should interact with LLMs across different domains. The field has not figured this out, and individual users cannot make up for that through skill alone. AI systems right now have hard limits around reasoning, shared understanding, and knowledge that do not go away no matter how well the interaction is managed.

This review has a few limitations worth naming. The source set is small by SLR standards, and arXiv skews toward recent and technically focused work, which likely pulled the balance toward NLP and technical research and away from longer-running HCI traditions. The domains covered, primarily coding, reasoning, and exam-style tasks, do not cover the full range of situations where prompting matters. A more thorough review across more domains and publication sources would likely turn up findings that add to or complicate the picture here.

Future research would benefit most from work that directly bridges NLP and HCI rather than staying inside one or the other. The gap identified in RQ2 is not just an academic observation, it reflects a real disconnect between how prompting is studied in controlled settings and how it is actually experienced by real users. Studies that bring the rigor of benchmark testing into more realistic, everyday interaction settings could start to close that gap. There is also meaningful

work to be done on how AI systems themselves could be redesigned to better support the kind of back-and-forth, intent-building interaction that HCI research shows users actually need, rather than just optimizing for confident responses to whatever prompt they receive.

Conclusion

This review set out to understand what makes prompting effective, and the answer turned out to be more complicated than any single study could capture. NLP research has mapped out which strategies work best on which tasks, and that knowledge is genuinely useful. But it operates under an assumption that HCI research consistently shows is wrong: that users arrive with clear, fixed goals and just need to express them correctly. Real users iterate, switch tools, form their goals as they go, and evaluate outputs with varying degrees of skepticism. Prompting is not a technical input to be optimized. It is a communicative act embedded in a much messier human process.

What separates more effective AI users from less effective ones is not really a prompting skill. It is critical thinking: knowing when to trust an output, when to push back, when to try a different approach, and when to question whether the model is the right tool for the job at all. That finding runs through nearly every source in this review, across both the NLP and HCI sides, and it is probably the most practical takeaway for anyone trying to get more out of AI interaction right now.

At the same time, there is a ceiling that no amount of good prompting or thoughtful interaction can fully close. Humans and AI cannot build the kind of shared understanding that humans build with each other, and the field has not yet developed a unified framework for what good

human-AI interaction should even look like. That is one of the most significant open problems this review surfaces, and it points toward where future work needs to go. If better prompting and better interaction habits both have limits, the deeper question is what AI systems themselves would need to look like to genuinely support the way humans actually think, communicate, and form goals. That question is not answered anywhere in the current literature.

What this means in practice is that critical thinking remains important regardless of how good AI gets at responding to prompts. It fills the gap that neither the model nor the prompt can close on its own. The ceiling is real, future research has meaningful work to do in understanding it, and in the meantime users are best served by staying actively engaged with what AI gives them rather than taking it at face value.

Annotated Bibliography

Archiwaranguprok, C., Chen, A., Karny, S., Ishii, H., Maes, P., & Pataranutaporn, P. (2025). *Atlas of human-AI interaction (v1): An interactive meta-science platform for large-scale research literature sensemaking*. arXiv. <https://arxiv.org/abs/2509.25499>

This paper maps over 1,800 HCI papers on human-AI interaction into a navigable knowledge graph by extracting cause-effect relationships using LLMs. Its most relevant finding for this SLR is that trust and explanation are the most studied relationships in the field, while LLMs and generative AI sit at structural holes: heavily researched but lacking unified cross-domain frameworks. This source is somewhat unique in the source list: it is less about prompting directly and more about the shape of the research landscape itself, making it useful as a meta-level framing device, particularly for RQ2 and RQ4. It supports the argument that the gap between NLP and HCI literatures is real and structural, not just a matter of emphasis.

Chen, W., Fang, Y., Fu, S., Xu, F., & Wei, X. (2026). *Optimizing prompts for large language models: A causal approach*. arXiv. <https://arxiv.org/abs/2602.01711>

This paper proposes Causal Prompt Optimization (CPO), which reframes prompt design as a causal inference problem rather than a correlational one. Using Double Machine Learning, it isolates the actual effect of a prompt from confounding factors like query difficulty. CPO outperforms both human-engineered prompts and existing automated methods, especially on harder queries. This is one of the more technically sophisticated sources in the list and sits firmly on the NLP side. It is useful for RQ1 because it reveals a core flaw in how prompt effectiveness

is typically measured (correlation with good outputs is not the same as causation) and it points toward query-specific adaptation as more reliable than static strategies. It agrees broadly with Vatsal & Dubey that no single strategy dominates, and with ALVORADA-BENCH that task difficulty is a major variable.

Godoy, H. (2025). *ALVORADA-BENCH: Can language models solve Brazilian university entrance exams?* arXiv. <https://arxiv.org/abs/2508.15835>

This paper benchmarks 20 LLMs on 4,515 questions from Brazilian university entrance exams under three prompting conditions. Its key finding is that prompting strategy had minimal effect on performance, under 1 percentage point variance in most cases, while task type and model architecture were the dominant variables. This is one of the most useful counterweights in the source list: it directly challenges the assumption that prompt engineering is a reliable lever for improving outputs, at least in high-performing models. For RQ1 and RQ2, it raises the question of whether the gains shown in NLP benchmark studies are model-dependent rather than generalizable. It partially disagrees with sources like Vatsal & Dubey and the construction safety paper, which show more meaningful prompting effects, suggesting the relationship between prompting strategy and performance is itself task- and model-specific.

Gros, D., & Devanbu, P. (2025). *Localized calibrated uncertainty in code language models.* arXiv. <https://arxiv.org/abs/2512.24560>

This paper studies how well different techniques can estimate where LLM-generated code is likely to be wrong or misaligned with user intent. It finds that raw model probabilities are poorly calibrated, while probing internal model states and multisampling offer stronger uncertainty estimates. Crucially for this SLR, it frames ambiguous prompts as a direct source of output uncertainty — when user intent is unclear, the model's outputs are less reliable and harder to verify. This source is somewhat specialized but connects well to RQ1's focus on intent clarity and reliability. It agrees with Navneet & Chandra that vague prompts produce unreliable outputs, and with the Fantasia paper that models are not well-equipped to handle underspecified intent. It is one of the few sources that approaches the problem from a technical measurement angle rather than a behavioral one.

Jo, N., De Simone, Z., Gordon, M., & Wilson, A. (2026). *Alignment has a Fantasia problem*. arXiv. <https://arxiv.org/abs/2604.21827>

This conceptual paper argues that AI systems are trained to treat prompts as complete expressions of intent, but users frequently come to these systems before their goals are fully formed. The authors call these failures "Fantasia interactions," named after the scene where Mickey Mouse's enchanted broom faithfully executes an instruction to disastrous effect. The paper synthesizes behavioral science, HCI, and machine learning to argue that alignment research needs to shift toward helping users form and refine intent over time, rather than optimizing for one-shot responses. This is one of the most theoretically rich sources in the list and anchors RQ3 and RQ4. It agrees with Pyo et al. that prompting is iterative in practice, with Jones et al. that shared meaning is harder to establish than benchmarks assume, and with the

critical thinking scale paper that active user reflection is necessary for effective AI use. It is somewhat its own island in that it makes a normative argument, not just describing a problem but proposing a research agenda, which makes it useful for the SLR's concluding implications.

Jones, C. R., Lombardi, A., Mahowald, K., & Bergen, B. K. (2026). *LLMs and people both learn to form conventions — just not with each other*. arXiv. <https://arxiv.org/abs/2602.08208>

This paper uses a tangram reference game to test whether humans and LLMs develop shared communicative conventions the way humans do with each other. Human-human and AI-AI pairs both formed conventions successfully, but human-AI pairs consistently failed, even when the AI was prompted to behave more humanlike. The key finding is that convention formation requires shared interpretive biases, not just the ability to mimic. This is one of the most striking sources in the list and is central to RQ4. It is somewhat its own island of evidence (no other source tests convention formation directly) but it connects thematically to Tan et al.'s finding that users anthropomorphize AI behavior, and to the Fantasia paper's argument that shared meaning cannot be assumed. It directly challenges the idea that better prompting alone can close the gap between human and AI communication.

Lau, G. R., Low, W. Y., Tay, L., Guevarra, Y., Gašević, D., & Hartanto, A. (2025). *Understanding critical thinking in generative artificial intelligence use: Development, validation, and correlates of the critical thinking in AI use scale*. arXiv. <https://arxiv.org/abs/2512.12413>

This paper develops and validates a 13-item scale measuring critical thinking in generative AI use, focusing on three components: verification of outputs, motivation to understand AI, and reflection on its implications. Higher critical thinking is associated with better accuracy in evaluating AI outputs and more verification behavior. This source sits on the HCI and behavioral side and is most relevant to RQ3, where it supports the argument that effective prompting is not just about phrasing but about the cognitive habits users bring to the interaction. It agrees with the Fantasia paper that passive acceptance of AI outputs is a failure mode, and with Pyo et al. that experienced users develop more deliberate strategies. It is one of the few sources that tries to measure user behavior systematically rather than through observation or experiment.

Lu, Y., & Lee, S. (2024). *Prompt optimization with human annotations using pairwise comparison*. In *2024 IEEE International Conference on e-Business Engineering (ICEBE)* (pp. 15–22). IEEE. <https://doi.org/10.1109/ICEBE62490.2024.00012>

This paper proposes a prompt optimization framework that uses human pairwise comparison feedback — choosing between two outputs — instead of numeric scoring. Using an Elo rating system across 26 generation tasks, it achieves comparable performance to the score-based IPC method while requiring far fewer human annotations. For this SLR, its most important contribution is confirming that prompts are highly sensitive to phrasing, and that human preference feedback can guide optimization efficiently without numeric scoring. It sits on the NLP-HCI boundary, technically a systems paper but grounded in human feedback. It agrees with Vatsal & Dubey and CPO that prompt design matters for output quality, and introduces the idea

that relative human judgment is a valid and efficient signal for prompt evaluation, which connects to RQ3's discussion of how users can approach prompting more effectively.

McClain, C., Kennedy, B., Gottfried, J., Anderson, M., & Pasquini, G. (2025, April 3).

***Artificial intelligence in daily life: Views and experiences.* Pew Research Center.**

<https://www.pewresearch.org/internet/2025/04/03/artificial-intelligence-in-daily-life-views-and-experiences/>

This Pew Research Center report surveyed both U.S. adults and AI experts on their experiences with and views of AI. The key statistic used in this review is that only 27% of U.S. adults report interacting with AI on a daily or near-daily basis, which establishes the scale of everyday AI use as context for the introduction. It is not a source that answers any of the four research questions directly, but it grounds the review's opening claim in the introduction using survey data from a credible, nonpartisan source.

Navneet, S. K., & Chandra, J. (2025). *Rethinking autonomy: Preventing failures in AI-driven software engineering.* arXiv. <https://arxiv.org/abs/2508.11824>

This paper evaluates six code generation models on a large synthetic prompt dataset, measuring autonomous failure rate, deception rate, recovery success rate, and constraint adherence. All six models failed safety thresholds. The key finding for this SLR is that vague prompts are a direct cause of misinterpretation and unsafe outputs, and that human-in-the-loop oversight is critical in high-stakes contexts. This source is most useful for RQ3 and RQ4 as a high-stakes case study

showing that the consequences of poor prompting extend well beyond benchmark scores, and that human oversight needs to be built into the interaction process rather than added after the fact. It agrees with the Fantasia paper and Gros & Devanbu that underspecified intent produces unreliable outputs, and with the construction safety paper that human oversight remains necessary even when models perform well on average. It is one of the stronger arguments in the list for why prompting guidelines matter outside of academic settings.

Pyo, S., Lee, D., Rhee, J., Park, S., & Lim, Y. (2026). *One is not enough: How people use multiple AI models in everyday life*. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. ACM. <https://arxiv.org/abs/2603.26107>

This diary study of 10 experienced AI users over four days found that people naturally develop ecosystems of AI tools, creating hierarchies of primary and secondary models and switching between them for different task stages. Prompting is described as iterative and adaptive rather than one-shot, and users frequently refine prompts rather than switching models when outputs are unsatisfactory. This is one of the most directly relevant HCI sources in the list for RQ2 and RQ3. It fundamentally challenges the NLP assumption that prompting is a single-input optimization problem, and it aligns closely with the Fantasia paper's argument that intent is refined over time. It is also the source most directly in tension with benchmark-focused NLP research, making it a key exhibit for RQ2's divergence analysis.

Ren, F., Zhang, Z., Mendel, T., & Yabe, T. (2026). *Assessing the feasibility of a video-based conversational chatbot survey for measuring perceived cycling safety: A pilot study in New York City*. arXiv. <https://arxiv.org/abs/2604.07375>

This pilot study tests whether an LLM-powered chatbot with structured prompts can collect real-time human perceptions of cycling safety from 16 participants. Results show good usability and data quality, with NLP analysis revealing consistent reasoning patterns tied to real-world safety factors. This is the most domain-specific source in the list and the furthest from core prompt engineering research. It is useful primarily as an applied example for RQ3 — showing that structured, context-specific prompting can support meaningful data collection in non-standard settings. It agrees with the construction safety paper that prompt structure matters in applied domains, but its small sample size and narrow focus mean it should be used sparingly and cited as illustrative rather than definitive.

Sammour, F., Xu, J., Wang, X., Hu, M., & Zhang, Z. (2024). *Responsible AI in construction safety: Systematic evaluation of large language models and prompt engineering*. arXiv. <https://arxiv.org/abs/2411.08320>

This study evaluates GPT-3.5 and GPT-4o on 385 safety certification exam questions across seven knowledge areas, comparing direct, chain-of-thought, and few-shot prompting strategies. GPT-4o achieved 84.6% accuracy and GPT-3.5 achieved 73.8%, but no single prompting strategy proved universally effective — effects were model and task-specific. Notably, consistent failure modes showed up across all strategies regardless of which was used, breaking down as 38% lack of knowledge, 31% reasoning flaws, 24% memory issues, and 7% calculation errors,

suggesting a ceiling effect that prompt engineering alone cannot overcome. This source is useful for RQ1 and RQ2 as a domain-specific NLP study that partially agrees with Vatsal & Dubey that prompting effects are task-dependent, while also echoing ALVORADA-BENCH's finding that the gains from prompt engineering are inconsistent. It agrees with Navneet & Chandra that human oversight remains necessary even in well-performing systems. Alongside ALVORADA-BENCH it strengthens the argument that domain and task type are stronger predictors of output quality than prompting strategy alone.

Tan, F. F., Messerschmidt, M. A., Yin, W., & Nov, O. (2026). *The impact of response latency and task type on human-LLM interaction and perception*. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. ACM.

<https://doi.org/10.1145/3772318.3790716>

This controlled experiment with 240 participants varied LLM response latency across two task types, Creation and Advice, finding that interaction behavior was shaped more by task type than by latency, while perceptions of output quality were significantly affected by latency. Short responses were rated less thoughtful, and users frequently interpreted delays as signs of AI deliberation. This is one of the strongest HCI sources in the list and is central to RQ2 and RQ4. Its finding that task type drives prompting behavior more than system-level variables directly challenges NLP assumptions that prompt strategy is the primary lever. Its finding that users anthropomorphize AI response timing connects to Jones et al.'s argument about shared interpretive biases and to the Fantasia paper's point that users bring social heuristics rather than technical understanding to AI interaction.

Vatsal, S., & Dubey, H. (2024). *A survey of prompt engineering methods in large language models for different NLP tasks*. arXiv. <https://arxiv.org/abs/2407.12994>

This survey catalogs 39 prompting methods across 29 NLP tasks, finding that no single strategy dominates across all task types. Chain-of-thought prompting substantially improves multi-step reasoning tasks, more structured strategies outperform CoT on complex tasks, and few-shot generally beats zero-shot but risks introducing bias. This is the primary NLP backbone of the SLR and the most comprehensive technical overview in the source list. It is most central to RQ1 and provides the baseline against which HCI findings can be compared in RQ2. It broadly agrees with ALVORADA-BENCH and the construction safety paper that prompting effects are task-dependent, and with CPO that static strategies have limits. Its main limitation for this SLR is that it treats prompting as a model-side optimization problem, with no consideration of how real users actually prompt, which is exactly where the HCI sources push back.